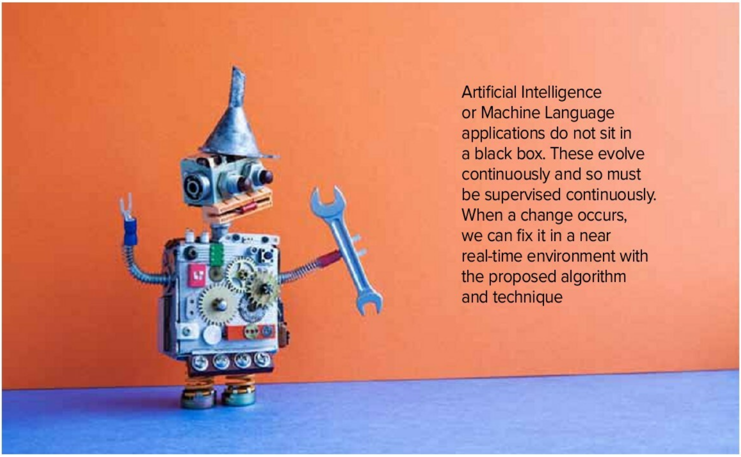
AI: The Curious Case Of Concept Drift



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Think about a scenario where you are using an electric motor on your shop-floor. You have installed a few sensors on this motor's outer body, and these sensors are continuously sending the data over the wireless network. Moreover, you also have an elaborate setup on the cloud where your application analyses these parameters and determines your motor's health status.

Of course, this scenario considers the trained pattern of motor vibration and current consumption by the motor during the learning phase. Moreover, so long as nothing changes, this pattern recognition works like a charm. That is what a good machine learning outcome would look like.

However, the data generated by an electric motor can change over time. It can result in poor analytical results, which otherwise assumes a static relationship between key parameters and motor-health.

The change in data occurs due to various real-life scenarios. These scenarios range from changes in operating load conditions, aging of mechanical components such as ball-bearings, or wear and tear of the foundation on which the motor is installed. Environmental conditions can change, and several other factors may get affected too.

Nonetheless, this occurrence is quite common for several other real-life scenarios